DSAA3073: Theories in Data Science

Week 8 Validation Sheet · Margins, Surrogates, and Evidence Boundaries

Coverage: Week 8 **Duration:** 30 minutes · **Total:** 130 marks

Answer all questions. For multiple-choice questions, select exactly one answer. Show the essential reasoning requested; unsupported answers may receive no credit.

Part A: Multiple choice

A1 [5 marks]. Let $F_\alpha(x) = \sum_t \alpha_t h_t(x)$ with $\alpha \neq 0$. Which quantity is unchanged when α is replaced by $c\alpha$ for any $c > 0$ and still records whether (x, y) is classified correctly?

(a) $yF_\alpha(x)$
(b) $|F_\alpha(x)|$
(c) $\frac{yF_\alpha(x)}{\|\alpha\|_1}$
(d) The Euclidean distance from the original input x to a decision boundary

Solution. (c). Under $\alpha \to c\alpha$, both $F_\alpha(x)$ and $\|\alpha\|_1$ are multiplied by the same positive factor c, so their signed ratio is unchanged. Its sign is positive on a correctly classified example and nonpositive on an error or tie. The raw quantities in (a) and (b) change scale. Option (d) is not determined without an input-space metric and decision-boundary representation.

Marking. *5 marks for (c) together with the cancellation of the common positive scale and the role of the sign; 0 otherwise.*

A2 [5 marks]. Which statement about $\hat{R}_{S,\theta} = m^{-1} \sum_i \mathbf{1}[\rho_i \leq \theta]$ and the Week 8 margin certificate is correct?

(a) $\hat{R}_{S,\theta}$ must decrease as θ increases.
(b) The displayed theorem fixes one positive θ before its probability statement.
(c) $\hat{R}_{S,\theta}$ is an observed test-error curve.
(d) Uniformity over normalized ensemble functions automatically permits choosing any θ after seeing the sample.

Solution. (b). The taught theorem fixes a positive threshold before drawing the sample and taking the high-probability event. The empirical low-margin fraction is a cumulative training quantity, so it is nondecreasing in θ, not decreasing. Uniformity over functions and uniformity over thresholds are different quantifiers; the former does not supply the latter.

Marking. *5 marks for (b) and the fixed-threshold quantifier; 0 otherwise.*

A3 [5 marks]. From $\mathbf{1}[yF(x) \leq 0] \leq e^{-yF(x)}$, which conclusion follows on a fixed training sample without any further theorem?

(a) Empirical zero-one error is at most empirical exponential loss.
(b) Population error equals empirical exponential loss.
(c) Exponential-loss minimization is robust to every type of label noise.
(d) Every optimizer of exponential loss follows AdaBoost's coefficient path.

Solution. (a). Averaging the pointwise inequality over the fixed training sample gives the stated empirical comparison. Population risk requires a separate generalization theorem, robustness requires data- and class-dependent evidence, and a shared objective does not determine one optimization path.

Marking. *5 marks for (a) and recognition that the consequence remains empirical; 0 otherwise.*

A4 [5 marks]. At the current score, suppose the directional derivative along coordinate h_k is $C_t\big(2\varepsilon_{t,k} - 1\big)$, where $C_t > 0$. Which claim is licensed?

(a) The coordinate with the smallest current weighted error has the most negative derivative.
(b) The common factor C_t may be declared equal to one in the derivative identity.
(c) Choosing the steepest coordinate proves the fitted model has the lowest test error.
(d) Any coordinate schedule therefore reproduces AdaBoost's entire path.

Solution. (a). Because C_t is positive and common across coordinates, ordering the weighted errors orders the derivatives. The factor may be ignored for this comparison but is not generally equal to one. A local empirical direction does not determine an entire optimization path or observed test behavior.

Marking. *5 marks for (a), including why the positive common factor preserves the ordering; 0 otherwise.*

Part B: Definition in use

B1 [20 marks]. A binary ensemble has coefficient vector $\alpha = (2, -1, 1)$ and, at one input, weak-hypothesis-output vector $h(x) = (1, -1, -1)$. The label is $y = -1$.

Compute $F_\alpha(x)$, $\|\alpha\|_1$, the normalized functional margin, and the unsigned L^∞ distance from $h(x)$ to the hyperplane $\{z : \alpha \cdot z = 0\}$. Repeat the margin calculation after replacing α by 3α. Then explain precisely why neither computed quantity is an input-space geometric distance and why the signed margin and unsigned output-space distance have different meanings on this example.

Solution. The real ensemble score is the coefficient-vector dot product with the weak-hypothesis outputs:

$$F_\alpha(x) = 2(1) + (-1)(-1) + 1(-1) = 2.$$

Its coefficient norm is

$$\|\alpha\|_1 = |2| + |-1| + |1| = 4.$$

Therefore the registered normalized functional margin is

$$\rho(x, y) = \frac{yF_\alpha(x)}{\|\alpha\|_1} = \frac{(-1)(2)}{4} = -\frac{1}{2}.$$

Its negative sign correctly records that $\operatorname{sign}(F_\alpha(x)) = +1$ does not equal the label -1.

Replacing α by 3α changes the score to 6 and the coefficient norm to 12. Hence the normalized functional margin remains $(-1)\frac{6}{12} = -\frac{1}{2}$. This explicitly verifies positive scale invariance.

In weak-hypothesis-output space, the dual-norm distance to the α-normal hyperplane is

$$d_\infty(h(x), \{z : \alpha \cdot z = 0\}) = \frac{|\alpha \cdot h(x)|}{\|\alpha\|_1} = \frac{|2|}{4} = \frac{1}{2}.$$

This quantity is unsigned. It records the distance of the point $h(x)$ in $\mathbb{R}^3$ from the output-space hyperplane, whereas the normalized functional margin attaches the label and is negative here. Neither quantity is a distance from the original input x: the prompt gives no metric on the input space, no feature-space geometry for x, and no input-space decision boundary. Equal magnitudes on this misclassified example therefore do not make the two objects interchangeable.

Marking. *4 marks for $F_\alpha(x) = 2$ and $\|\alpha\|_1 = 4$; 4 for $\rho = -\frac{1}{2}$ and its misclassification interpretation; 4 for the rescaled score, norm, and unchanged margin; 4 for the unsigned output-space distance $\frac{1}{2}$ with the correct point and hyperplane; 4 for explaining that no input-space distance is defined and distinguishing signed confidence from unsigned output-space distance.*

Part C: Basic skills

C1 [7 marks]. Ten normalized training margins are

$$-0.10, 0.05, 0.20, 0.20, 0.35, 0.55, 0.60, 0.80, 0.90, 1.00.$$

At threshold $\theta = 0.20$, compute $\hat{R}_{S,\theta}$. If the two nonempirical terms in the fixed-threshold bound are 0.18 and 0.07, compute the resulting population-error upper bound.

Solution. The cumulative definition counts every margin at or below the threshold. The qualifying values are $-0.10, 0.05, 0.20, 0.20$, so

$$\hat{R}_{S,0.20} = \frac{4}{10} = 0.40.$$

Adding the distinct complexity and confidence terms gives

$$R(\overline{F}) \leq 0.40 + 0.18 + 0.07 = 0.65.$$

This is a conditional population upper bound under the theorem's class, i.i.d.-sample, fixed-threshold, and confidence conditions; it is not an observed error or an equality.

Marking. *4 marks for the at-or-below count and $\hat{R} = 0.40$; 2 for the bound 0.65; 1 for interpreting it as a conditional upper bound rather than an observed equality.*

C2 [8 marks]. A fitted score has empirical exponential loss 0.18. At one coordinate step, $C_t = 0.24$ and the current weighted error is $\varepsilon_{t,k} = 0.35$. Give the empirical zero-one-error conclusion and compute the directional derivative $C_t\big(2\varepsilon_{t,k} - 1\big)$. Interpret its sign.

Solution. Averaging the pointwise surrogate inequality yields

$$\hat{R}_{0-1}(F) \leq \hat{L}_{\exp}(F) = 0.18.$$

This conclusion concerns the same training sample. The derivative is

$$0.24(2(0.35) - 1) = 0.24(-0.30) = -0.072.$$

Because it is negative, a sufficiently small positive move along this coordinate decreases the empirical exponential objective locally. This derivative alone is not a population or held-out-performance claim.

Marking. *3 marks for the empirical zero-one bound ≤ 0.18; 3 for substituting into the derivative and obtaining -0.072; 2 for the local-descent interpretation and its empirical limitation.*

Part D: Construction and diagnosis

D1 [17 marks]. On two inputs a, b, let

$$(h_1(a), h_1(b)) = (1, 1), \quad (h_2(a), h_2(b)) = (1, -1).$$

Consider the signed-coefficient normalized score $G = \frac{2h_1 - h_2}{3}$.

Use these two input values to prove that G is not in $\mathrm{conv}(\{h_1, h_2\})$ under the Week 8 convention of nonnegative weights whose sum is exactly one. Then construct a valid convex-hull representation after making the dictionary sign-closed, and explain why division by an L^1 coefficient norm did not by itself license the original convex-hull claim.

Solution. First evaluate the signed score on the two inputs:

$$(G(a), G(b)) = \frac{1}{3}(2(1, 1) - (1, -1)) = \left(\frac{1}{3}, 1\right).$$

If G belonged to the convex hull of the original dictionary, there would be coefficients $\mu, \nu \geq 0$ with $\mu + \nu = 1$ such that $G = \mu h_1 + \nu h_2$. At input a, both base hypotheses equal one, so every such convex combination has value

$$\mu h_1(a) + \nu h_2(a) = \mu + \nu = 1,$$

whereas $G(a) = \frac{1}{3}$. This contradiction already proves nonmembership. The second input confirms the incompatibility: matching $G(b) = 1$ would require $\mu - \nu = 1$, and together with $\mu + \nu = 1$ this forces $(\mu, \nu) = (1, 0)$, which still gives value 1 rather than $\frac{1}{3}$ at a. Thus the two displayed input values form a complete counterexample to membership in the original convex hull.

If the dictionary is explicitly enlarged to be sign-closed, then $-h_2$ is an available base hypothesis and

$$G = \frac{2}{3}h_1 + \frac{1}{3}(-h_2).$$

These weights are nonnegative and sum to one, so this is a valid convex combination in the enlarged dictionary.

The denominator $3 = |2| + |-1|$ removes a common positive scale, but it does not change the sign of the coefficient on h_2. Consequently it creates absolute weights that sum to one, not nonnegative weights over the original hypotheses. Sign closure or an explicit reparameterization is the missing class condition.

Marking. *4 marks for evaluating G as $\left(\frac{1}{3}, 1\right)$; 6 for enforcing $\mu + \nu = 1$ and deriving the value-at-a contradiction, or an equivalent two-input inconsistency; 4 for the valid sign-closed representation $2\frac{h_1}{3} + \frac{-h_2}{3}$; 3 for explaining why L^1 normalization removes scale but does not create nonnegative weights in the original dictionary.*

D2 [18 marks]. During several boosting rounds, one training example's normalized weight rises from 0.03 to 0.41 while its signed score remains negative. No label audit or held-out results are available.

Construct two distinct data-generating or weak-class explanations compatible with this same trace: one in which the recorded label is wrong and one in which it is correct. For each explanation, identify evidence that could distinguish it from the other. Finally state exactly what the weight trace establishes and what it cannot establish about label correctness and test performance.

Solution. The update mechanics explain the observed concentration. A negative signed score makes $e^{-y_i F(x_i)}$ large relative to easier examples, and normalization can therefore raise this example's share of the total weight. That mechanism depends on current fitted scores, not on direct knowledge of whether a recorded label is correct.

One compatible world contains a labeling error: the observed feature pattern belongs to a region whose verified class is $+1$, but the training record was accidentally entered as -1. Weak hypotheses that fit the surrounding clean pattern repeatedly disagree with the recorded label, so its weight rises. Evidence could come from checking the original annotation, obtaining an independent expert relabel, or comparing repeated high-quality labels.

A second compatible world has a correct but rare example. Its class is truly -1, yet the declared weak-hypothesis class cannot isolate its local region, or the learner is still at an early stage. The available weak rules keep predicting $+1$, producing almost the same trace. Evidence could come from a trusted label audit confirming -1, from enlarging the weak class under a predeclared comparison and seeing whether the example becomes representable, or from examining performance on independently labeled examples from the same region.

Therefore the trace establishes persistent current difficulty and increasing relative influence under the exponential reweighting mechanism. It does not identify a mislabeled point, prove a universal noise threshold, or determine population or observed test error. Test-performance claims need a valid margin certificate under all its conditions or, for the fitted model's observed behavior, untouched held-out evidence.

Marking. *4 marks for linking the rising normalized weight to persistent negative/small signed score under exponential reweighting; 4 for a coherent mislabeled-example world; 4 for a coherent correctly labeled but weak-class-limited or rare-region world; 3 for evidence capable of distinguishing the worlds; 3 for the exact limitations on label diagnosis and test-performance conclusions.*

Part E: Proof and derivation

E1 [25 marks]. For each $\theta \in \{0.20, 0.50\}$ fixed before sampling, a current theorem states that, with probability at least $1 - \eta$, simultaneously for every $\overline{F}$ in the normalized ensemble class,

$$R(\overline{F}) \le \hat{R}_{S,\theta}(\overline{F}) + \frac{0.04}{\theta} + \sqrt{\frac{\log\left(\frac{1}{\eta}\right)}{400}}.$$

After seeing one fitted ensemble's training-margin curve, an analyst obtains $\hat{R}_{S,0.20} = 0.08$ and $\hat{R}_{S,0.50} = 0.30$. The analyst uses $\eta = 0.10$ separately for both candidates, selects the smaller right-hand side, and reports a 90% confidence certificate.

Audit this report and construct a numerical certificate that legally permits selection after both empirical values are seen while retaining overall confidence at least 90%. State the probability event and order of choices.

Available fact. $\Pr(\cup_j B_j) \le \sum_j \Pr(B_j)$ for finitely many bad events. Use $\log(10) \approx 2.303$ and $\log(20) \approx 2.996$.

Solution. The theorem is uniform over the ensemble $\overline{F}$ only after one threshold has been fixed. The fitted ensemble may therefore be selected from the normalized class on the same sample, but the displayed statement does not automatically make the confidence event simultaneous over the two thresholds. The analyst inspected both empirical margin values and then selected θ, so a one-threshold event at failure probability 0.10 is not enough.

Indeed, if each threshold is separately assigned failure probability 0.10, the available finite-union fact bounds the probability that at least one of the two fixed-threshold statements fails by 0.20. The event on which both candidates are safe is then guaranteed only at level 0.80, not 0.90. Numerically, the analyst's common confidence term would be

$$\sqrt{\frac{\log(10)}{400}} \approx 0.0759,$$

producing candidate values 0.3559 and 0.4559. The lower arithmetic value does not repair its probability scope.

A legal repair is to declare the two candidate thresholds and allocate their failure probabilities **before drawing the sample**. For example, set $\eta_{0.20} = \eta_{0.50} = 0.05$. Let B_θ be the bad event that the displayed inequality fails for at least one normalized ensemble at that fixed threshold. Each theorem instance gives $\Pr(B_\theta) \le 0.05$, and hence

$$\Pr(B_{0.20} \cup B_{0.50}) \le 0.05 + 0.05 = 0.10.$$

Therefore, with probability at least 0.90, both fixed-threshold statements hold simultaneously for every normalized ensemble in the class.

On this intersection event the shared confidence term is

$$\sqrt{\frac{\log(20)}{400}} \approx \sqrt{0.00749} \approx 0.0865.$$

The two valid simultaneous candidates are

$$\theta = 0.20: \quad 0.08 + \frac{0.04}{0.20} + 0.0865 = 0.3665,$$

and

$$\theta = 0.50: \quad 0.30 + \frac{0.04}{0.50} + 0.0865 = 0.4665.$$

The analyst may now inspect both empirical values and choose $\theta = 0.20$, reporting

$$R(\overline{F}) \le 0.3665$$

with overall confidence at least 90%, subject to the stated i.i.d. sample and normalized-class conditions. More generally, any predeclared allocation $\eta_{0.20} + \eta_{0.50} \le 0.10$ yields a valid simultaneous event. The allocation, candidate set, and computation rule must be fixed before the data; only the choice among the resulting simultaneously certified candidates may occur after seeing the margin curve.

Marking. *5 marks for identifying that post-sample threshold selection requires an event covering both candidates and that function-uniformity does not supply threshold-uniformity; 5 for translating the selected report into an intersection of two fixed-threshold events and showing why two separate* $\eta = 0.10$ *uses give only an* 0.80 *guaranteed intersection; 5 for predeclaring a valid failure allocation whose total is at most* 0.10*; 5 for deriving the simultaneous-event probability and the corrected candidate certificates* 0.3665 *and* 0.4665*; 5 for legally selecting* $\theta = 0.20$ *only after the simultaneous event is established and stating the final 90% certificate with its remaining class/sample conditions.*

Part F: Synthesis and limitations

F1 [15 marks]. A report about one fitted boosting model contains four sentences:

1. Its empirical exponential loss is 0.06, so its empirical zero-one error is at most 0.06.
2. Any other optimizer reaching objective value 0.06 must have followed the same AdaBoost path and produced the same classifier.
3. A valid fixed-threshold margin bound has right-hand side 0.24, so the model's observed test error equals 0.24.
4. An untouched held-out set gives error 0.11, so every future sample and every boosting problem must have error 0.11.

For each sentence, identify the relevant evidence type: objective value, AdaBoost update path, margin certificate, or held-out performance. Then write the strongest conclusion that the evidence actually supports. State any missing conditions needed for the stronger claims.

Solution. Sentence 1 is the one licensed direct consequence of the **objective value**. Pointwise domination and averaging on the same training sample give $\hat{R}_{0-1} \le \hat{L}_{\exp} = 0.06$. It is an empirical inequality, not a population conclusion.

Sentence 2 confuses an objective value with an **update path**. The exact AdaBoost path requires the declared greedy minimum-current-weighted-error coordinate rule, the appropriate binary dictionary or oracle conditions, its boundary conditions, and the matching step rule. A different optimizer can attain the same scalar loss with different coefficients, margins, signs, or iterates. Equality of objective values alone supports none of those equalities.

Sentence 3 concerns a **margin certificate**. Under its normalized-class, i.i.d.-sample, fixed-threshold, base-class-complexity, and confidence conditions, the theorem says population zero-one error is at most 0.24 on its high-probability event. It is an upper bound, not an equality, and it is not an observed test measurement.

Sentence 4 uses **held-out performance**. If the held-out data were untouched by fitting, threshold choice, stopping, and reporting decisions, 0.11 is an estimate of the fitted procedure's performance under the held-out sampling regime. Sampling uncertainty and distributional scope remain. One observed set cannot prove an identical error on every future sample or a universal theorem about all boosting problems.

Thus objective value, path mechanics, a conditional margin theorem, and held-out evidence answer four different questions. None automatically supplies the others' conclusion.

Marking. *4 marks for validating sentence 1 only as an empirical surrogate upper bound; 4 for rejecting sentence 2 and naming the additional algorithmic path conditions; 4 for interpreting* 0.24 *as a conditional high-probability upper bound rather than observed equality; 3 for interpreting* 0.11 *as held-out evidence with sampling/distributional limits rather than a universal guarantee.*